

# Technical White Paper: The Empirical Ballistic Measurement Station (EBMS) v2.0

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## 1. Executive Summary

The EBMS is a forensic-grade ballistic measurement platform designed to eliminate the "black box" algorithmic estimations common in Doppler radar. By utilizing an **Asymmetric Multi-Node Architecture** and **TCXO-Stabilized Coherent Synchronization**, the system provides a mathematically reversible, air-gapped record of projectile flight. All measurements are anchored to high-stability reference clocks and environmental sensors, ensuring empirical accuracy that exceeds laboratory-grade standards. Version 2.0 upgrades the signal chain to a **200MHz/5.0ns architecture**, replacing standard phototransistors with **High-Speed PIN Photodiodes** and **LM318 Transimpedance Amplifiers**.

## 2. Technical Architecture & Data Acquisition

### 2.1 Asymmetric Tiered Processing

The system utilizes four specialized computing tiers to ensure zero-jitter execution and real-time visualization:

- **Tier 1 (Muzzle Execution):** **RP2040** microcontroller driven by a **12MHz TCXO ( $\pm 2.0\text{ppm}$ )**. Utilizing **PIO (Programmable I/O)** state machines with **bypassed input synchronizers**, it achieves a certified **5.0ns resolution** at a 200MHz system clock.
- **Tier 2 (System Orchestration & HMI):** **Raspberry Pi 5** running a high-priority **Python environment**. Provides a deterministic environment for sensor fusion

and the **7" Touchscreen Interface**, rendering real-time target maps and projectile placement.

- **Tier 3 (Terminal Acquisition):** Secondary **TCXO-stabilized RP2040 node** utilizing four independent PIO channels for cycle-accurate **Acoustic Time-of-Arrival (ToA)** capture and optical trigger.
- **Tier 4 (HMI Pendant):** Decoupled **Arduino Nano Every** via **RS-232**, providing **"Big Digit"** rendering on a **20x4 LCD (4-bit mode)** and physical Start/Stop controls.

## 2.2 High-Saturation Analog Front-End (The "Optical Web")

Based on a high-density summing matrix architecture, the EBMS creates a continuous optical "web":

- **Sensor Density:** 36 IR Emitters (**Vishay TSHG6200**) and 16 High-Speed Receivers (**Vishay BPV10NF PIN Photodiodes**) per screen (108/48 total).
- **Summing Logic (The 318 Upgrade):** 16 receivers per screen are tied into an active **Virtual Ground Summing Node** using an **LM318N High-Speed Op-Amp** (70V/ $\mu$ s slew rate). This converts photodiode current to voltage instantly, bypassing the microsecond-scale lag of traditional phototransistors.
- **Precision Thresholding:** Signals are fed to **TLV3501 ultra-high-speed comparators (4.5ns delay)**, providing a clean digital edge for the 5.0ns PIO clock.

## 2.3 Triple-Redundant Operational Logic

The EBMS enforces a "Triple-Check" timing protocol, simultaneously capturing:

1. **Segment 1-2** | 2. **Segment 2-3** | 3. **Segment 1-3 (Master)**

By cross-referencing all three segments, the system identifies and rejects

non-conforming data (muzzle blast, debris) that would compromise standard chronographs.

### 3. Physical Integrity & Synchronization

#### 3.1 Isochronous Signal Propagation

To maintain timing integrity, the EBMS enforces a strict physical layer:

- **Matched-Length Interconnects:** All screens must use identical-length, **Shielded Twisted Pair (STP)** cables (e.g., 10ft Cat7). This ensures propagation delay is a constant across all channels, eliminating phase-shift errors.
- **Deterministic Guardrails:** The Python logic monitors for cable-induced latency; mismatched or compromised cabling will trigger a **"Sync Error"** on the 7" Touchscreen and lock out data acquisition.

#### 3.2 Coherent Optical Handshake (The "Sync Loop")

- **Bidirectional Laser Link:** A 2-way 5mW green laser pulse provides a light-speed **"Time Zero"** trigger between the Muzzle (Tier 1) and Terminal (Tier 3) nodes.
- **Phase Locking:** This handshake cancels out propagation delays across the range, ensuring both TCXO-stabilized nodes are synchronized within a 5.0ns window.

#### 3.3 Mechanical Damping & Calibration

- **Chassis Construction:** Constructed from **sand-filled conduit**, utilizing high-mass damping to isolate sensors from muzzle blast "ringing." Transducers are further isolated via rubber mounts.

- **Parallelism Check:** The chassis ensures three optical curtains remain perfectly parallel. Users verify center-to-center spacing to the sub-millimeter (distance "d"), entered via the HMI Pendant.
- **Remote Terminal Acquisition:** The Tier 3 node is tripod-mounted and decoupled from the impact zone via tethered acoustic arrays. The Tier 2 Python layer applies an **"Electronic Offset"** to account for the fixed propagation delay of these tethered cables, ensuring **Hyperbolic Triangulation** remains centered on the physical target plane.

## 4. Security, Traceability & Data Integrity

### 4.1 Universal Traceability (The "Reconstruction Pathway")

The EBMS records **Raw Pulse-Counts (200MHz integer clicks)** from every node.

By retaining unfiltered clock cycles and environmental register data (**Bosch BMP280/BME280** at both Target and Control Console), the system creates a transparent audit trail for third-party verification.

### 4.2 Physical Non-Repudiation (QR Manifests)

Every shot string generates a physical manifest via an embedded **USB Thermal Printer**.

- **SHA-256 Hashing:** The manifest embeds a high-density QR Code containing the Raw Clicks, environmental influences, and a cryptographic hash.
- **Air-Gapped Security:** The system contains **no wireless transceivers** (No Wi-Fi/BT/Cellular), ensuring total operational security (OPSEC). Data export is via **USB Thumb Drive**.

### 4.3 Geospatial & Temporal Authentication

To ensure absolute non-repudiation, the EBMS integrates an **Active GPS/GNSS Node**:

- **Atomic Time-Stamping**: Every shot string is indexed to a GPS-derived UTC timestamp.
- **Geospatial Metadata**: Precise GPS coordinates and MSL (Mean Sea Level) altitude are captured.
- **Atmospheric Verification**: GPS-derived altitude is cross-referenced with the **BMP280** data at **both nodes** to validate air density profiles.

## 5. Setup & Calibration: The Isochronous Guardrail

### 5.1 Propagation Delay Matching

Because electrical signals travel through copper at roughly 8 inches per nanosecond, a mismatch of just 4 feet between cables would create a 6ns error—effectively "blinding" the system's high-resolution clock. The EBMS requires identical-length **Shielded Twisted Pair (STP)** cables to ensure propagation delay remains a constant that is mathematically cancelled during differential timing.

### 5.2 Deterministic Error Trapping (Python Logic)

The Tier 2 (Raspberry Pi/Python) layer performs a "Pre-Flight" timing validation. If the system detects a phase shift or rise-time variance between screens, the logic flags a **"Sync Error"** on the 7" Touchscreen and refuses to generate a SHA-256 Hashed QR Manifest, ensuring no compromised data enters the permanent record.

### 5.3 Mechanical Alignment & Plane Calibration

- **Parallelism Check:** The sand-filled conduit chassis ensures that the three optical curtains remain perfectly parallel.
- **Zero-Point Calibration:** Users verify the spacing between screen centers to the sub-millimeter. This physical distance ("d") is entered into the HMI Pendant, allowing the EBMS to calculate velocity with a precision that exceeds the measurement capabilities of standard Doppler radar.
- **Verification:** This ensures the Hyperbolic Triangulation remains centered on the physical target plane, regardless of the distance between the tripod-mounted sensors and the acquisition hardware.

#### 5.4 Dynamic Latency Compensation (HMI Config)

To accommodate varying range layouts, the Tier 2 Python environment includes a dedicated **Cable Calibration Interface**:

- **User-Defined Offsets:** The operator inputs the physical length of the tethered acoustic cables (e.g., 10ft, 15ft, 25ft) via the 7" Touchscreen.
- **Algorithmic Normalization:** The system automatically calculates the electrical propagation delay (based on the velocity of propagation of the matched STP cabling) and applies a real-time **"Time-of-Flight" correction** to the Tier 3 (Terminal Acquisition) node.

#### 5.5 Automated Self-Calibration & Warm-Up

To ensure peak metrological performance, the EBMS enforces a **120-second Initialization Protocol**:

- **Thermal Stabilization:** The system remains in a "Ready-Pending" state while the **TCXOs** reach a steady thermal equilibrium, eliminating frequency drift.

- **GPS Clock Disciplining:** Utilizing the **1 PPS GPS signal**, the system calculates the absolute frequency of the RP2040 clock, correcting for any sub-microsecond aging or temperature deviations.
- **Dynamic Thresholding:** During the warm-up, the Tier 2 Python tier analyzes the sensor noise floor, automatically adjusting the **TLV3501** comparator thresholds to maximize sensitivity while preventing false triggers from environmental interference.

## 5.6 Geospatial Lock & Atomic Synchronization

The EBMS requires a confirmed **GNSS/GPS 3D Fix** before entering an active firing state:

- **Temporal Integrity:** By anchoring the internal timing to the GPS atomic clock, the system ensures that the SHA-256 Hash is indexed to a non-repudiable universal time.
- **Geospatial Verification:** The system automatically captures the firing point's coordinates and altitude.
- **Atmospheric Validation:** GPS-derived altitude is cross-referenced with the **BMP280** Barometric Pressure at **both the Control and Terminal nodes**, providing a dual-source validation of the air density profiles.
- **Ready-State Interlock:** The READY indicator remains in a "Pending" state until thermal stabilization, GPS-lock, and 2-way laser sync are achieved.

## 5.7 The "Pre-Flight" Dashboard & System Health

During the 120-second initialization, the 7" Touchscreen provides real-time telemetry:

- **GNSS Satellite Tracking:** A live counter displaying "Sats Found" and signal strength.

- **Thermal Progress:** A visual status of the TCXO stabilization, ensuring the 200MHz clock is within the required  $\pm 2.0\text{ppm}$  tolerance.
  - **Sensor Health:** A rolling poll of the **dual BMP280/BME280** sensors and TLV3501 noise floor.
  - **Visual Handshake:** Once all parameters are met—thermal stability, GPS-lock, and 2-way laser sync—the READY indicator transitions to "Active," unlocking the SHOOT state.
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## EBMS Master Bill of Materials (BOM)

### 1. Processing & Timing (The "Logic Core")

- **[2x] RP2040 Microcontrollers:** (Tiers 1 & 3) For high-speed PIO pulse capture at 200MHz.
- **[1x] Raspberry Pi (4 or 5):** (Tier 2) Running **FreeRTOS** for orchestration and HMI.
- **[1x] 16GB+ Industrial eMMC Module:** For Tier 2 forensic data storage (High-reliability).
- **[2x] 12.000MHz TCXOs:** (e.g., Abracon FOX924B) High-stability timing reference ( $\pm 2.0\text{ppm}$ ).
- **[1x] u-blox Neo-M8 GNSS Module:** For **1 PPS Atomic Sync** and Geospatial metadata.
- **[1x] Arduino Nano/Pro Micro:** (Tier 4) Dedicated HMI controller for the "Big Digit" pendant.
- **[2x] DC-DC Buck Converters:** Adjusted to **1.15V** for RP2040 overclocking.

### 2. High-Saturation Optical Curtains (3 Screens)



- **[108x] High-Output IR Emitters:** (36 per screen) For the saturated overhead matrix.
- **[48x] High-Speed IR Photodiodes:** (16 per screen) For the **Active Summing Matrix**.
- **[3x] TLV3501 Comparator Modules:** Ultra-high-speed (**4.5ns delay**) for digital edge triggering.
- **[3x] LM224 Quad Op-Amps:** For the **Active Summing Node** conditioning logic.
- **[3x] Linear PCB Strips:** Custom or prototype boards for precise **1.1" sensor spacing**.

### 3. Forensic Sensing & Sync (100-Yard Link)

- **[2x] 5mW Green Laser Diodes:** (532nm) For the **2-way Coherent Sync** and "Time Zero."
- **[2x] BPW34 High-Speed Photodetectors:** For nanosecond laser pulse detection.
- **[4x] 12mm PZT Piezo Discs:** Acoustic sensors for the **Hyperbolic Triangulation** array.
- **[2x] BME280/SHT4x Sensors:** For dual-source environmental monitoring (Muzzle & Target).

### 4. HMI & Physical Output

- **[1x] 7" DSI Touchscreen:** Primary interface for the **Pre-Flight Dashboard**.
- **[1x] 20x4 Character LCD:** For "Big Digit" rendering on the Tier 4 Pendant.
- **[1x] 58mm Embedded Thermal Printer:** For physical, **SHA-256 Hashed QR Manifests**.
- **[1x] RS-232 Transceiver Set (MAX3232):** For noise-immune Pendant communication.

### 5. Mechanical & Connectivity (The "Isochronous Layer")

- **[Set] Matched-Length Shielded Twisted Pair (STP) Cables:** Specifically for the **8-inch-per-nanosecond** propagation delay matching.
  - **[6x] Amphenol Bayonet Connectors:** Mil-Spec style for vibration-proof, shielded interconnects.
  - **[2x] 12V Lithium-Ion Battery Packs:** Isolated power for Console and Target nodes.
  - **[Variable] 1" Conduit:** Chassis material (to be **filled with dry play sand** for damping).
  - **[Set] Rubber Isolation Mounts:** For PZT transducers and optical sensor strips.
  - **[2x] Spirit Levels:** For manual **Parallelism and Plane Calibration**.
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## EBMS Project: Legal Disclaimer & Open-Source Notice

### Notice of Non-Commercial Research & Intellectual Property Status

The following statement clarifies the intent of the developer and the non-commercial, metrological nature of this research.

#### 1. Experimental Intent & Non-Commercial Status

The **Empirical Ballistic Measurement Station (EBMS)** is an experimental work-in-progress developed by an independent researcher for private use, ballistic metrology research, and open-source community collaboration. The author has no commercial intent, does not offer this device for sale, and assumes no liability for the construction or use of similar devices by third parties.

#### 2. Intellectual Property & Public Domain Physics

The EBMS utilizes **Public Domain Physics** (Hyperbolic Triangulation, Speed of Sound/Light constants) and **Standard Laboratory Timing Techniques** (TCXO-disciplined PIO capture). Any similarity to existing commercial products is purely coincidental and a result of shared physical principles. The author maintains the right to document these "Garage-Built" innovations to push the boundaries of hobbyist-accessible forensic timing.

#### 3. Safety & High-Energy Warning

The EBMS involves high-energy projectiles and **Class 3R (5mW) Laser Systems**.

Users are solely responsible for ensuring:

- **Laser Safety:** Prevention of direct eye exposure to laser diodes.
- **Range Integrity:** Maintaining a clear and safe firing line.
- **Hardware Stability:** Proper anchoring of sand-filled conduit and tripods to prevent mechanical failure.

#### 4. Forensic Integrity Disclaimer

While the EBMS is designed for **Forensic-Grade Traceability** using **SHA-256 Hashing** and **GPS Atomic Sync**, the author makes no guarantees regarding the admissibility of this data in legal proceedings. The system is provided as a tool for empirical discovery and data-driven ballistic analysis.

#### 5. Open-Source Collaboration

The author welcomes input from the engineering community to ensure the EBMS remains a compliant, open-source resource. This project is dedicated to the spirit of **Empirical Truth** as championed by pioneers like Dr. Ken Oehler and P.O. Ackley.

This **EBMS Interface Map** summarizes the electrical and signal pathways

#### Tier 1: Screen Control (Tripod Node)

*The high-speed "Pulse Collector" and Laser Sync hub.*

- **Heartbeat:** 12.000MHz FOX924B TCXO injected into RP2040 XIN.
- **Power:** 12V Li-Ion
- **1.15V Buck** (Core) & **3.3V LDO** (Logic).
- **Inputs (3x Amphenol): Matched-Length Cat7 STP** carrying 5ns Digital Edges from screens.
- **Comms Link (Target): 532nm Laser** (Out) and **BPW34/TLV3501** (In) for "Time Zero" sync.
- **Data Link (Console): CAN Bus** (High/Low) twisted pair back to Tier 2.

## Tier 2: Control Console (Table Node)

*The "Forensic Brain" for HMI, metadata, and SHA-256 hashing.*

- **Heartbeat: Raspberry Pi 5** internal clock (Orchestrator).
- **Environment: Bosch BMP280 #1** (Console) + **Neo-M8 GNSS** (GPS Atomic Sync).
- **Comms Link (Tier 1): CAN Bus** twisted pair (Data from Tripod).
- **Output (Forensic): USB Thermal Printer** (QR Manifest) + **USB Thumb Drive** (Export).
- **HMI: 7" DSI Touchscreen** + **RS-232 Port** to HMI Pendant.

## Tier 3: Terminal Acquisition (Target Node)

*The "Terminal Ear" for acoustic triangulation and impact sync.*

- **Heartbeat: 12.000MHz FOX924B TCXO** injected into RP2040 **XIN**.
- **Environment: Bosch BMP280 #2** (Target) for dual-source atmospheric validation.
- **Inputs: 4x PZT Piezo Discs** (Acoustic ToA) via shielded cables.
- **Comms Link (Tripod): Laser Com Channel** (Handshake with Tier 1).

## Tier 4: HMI Pendant (Handheld Node)

*The "Human Interface" for rugged, non-real-time control.*

- **Heartbeat: Arduino Nano Every** (5V Logic).
- **Display: 20x4 LCD** (4-Bit Parallel Mode) for "Big Digit" rendering.
- **Controls:** Physical **Start/Stop Buttons** + Status LEDs.
- **Comms Link (Console): RS-232** ( $\pm 12V$ ) for high-noise-immunity serial data.

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## Signal Integrity & Power Matrix

| Path | Physical Layer | Critical Hardware Spec | Timing Goal |
|------|----------------|------------------------|-------------|
|------|----------------|------------------------|-------------|

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|                           |                           |                                                      |                    |
|---------------------------|---------------------------|------------------------------------------------------|--------------------|
| <b>Optical Screens</b>    | <b>Matched-Length STP</b> | 8-inches-per-nanosecond match                        | <b>5.0ns</b>       |
| <b>T1</b>                 |                           |                                                      |                    |
| <b>Laser Sync (T1 T3)</b> | <b>Light Link</b>         | <b>TLV3501</b> (4.5ns) / <b>BPW34</b> (2.5ns)        | <b>5.0ns</b>       |
| <b>Data (T1 T2)</b>       | <b>CAN Bus</b>            | Differential Twist / Ground Reference                | <b>N/A (Data)</b>  |
| <b>HMI (T4 T2)</b>        | <b>RS-232</b>             | <b>MAX3232</b> Transceiver                           | <b>Human-Scale</b> |
| <b>Power (All)</b>        | <b>3-State Switch</b>     | <b>Standby:</b> TCXO Stability / <b>On:</b> Emitters | <b>±2.0ppm</b>     |

<https://forms.gle/yUEqwwaM5G8ieqta9>